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Review Article

Artificial intelligence for decision-making in healthcare: A critical review of decision-maker-governed hybrid systems across clinical, operational, and policy settings

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ABSTRACT

Objectives: Artificial Intelligence (AI) in healthcare has advanced rapidly in prediction, classification, and pattern recognition, yet high-stakes clinical and operational decisions are not prediction problems alone. Clinical, operational and policy choices also require traceable reasoning, guideline fidelity, attention to institutional constraints, and accountability. This critical narrative review aims to synthesise evidence on neuro-symbolic and knowledge-infused AI for healthcare decision-making, with particular emphasis on decision-maker-governed systems: hybrid architectures in which clinicians, managers, or policymakers author, approve, or govern the symbolic knowledge structures that materially influence system behaviour.

Methods: We conducted a critical narrative review informed by a structured search of ten databases (last updated May 2026), screening 3,892 de-duplicated records to 101 sources retained for in-depth synthesis against three pre-specified analytic questions. The review proposes an operational definition for this category, distinguishes it from hybrid systems that merely import fixed external knowledge, and appraises evidence across clinical decision support, information extraction, hospital operations, and population-health governance.

Results: The literature suggests three cautious conclusions. Hybrid designs appear most credible in rule-bound, audit-sensitive, and workflow-dependent settings, including documentation validation, guideline-based recommendation, multimodal oncology workflows, and service optimisation under explicit constraints. Yet the empirical base remains limited; prospective evidence is sparse, comparisons between stakeholder-authored and non-stakeholder-authored hybrids are rare, and many claims regarding trust, fairness, and actionability remain largely theoretical. Symbolic layers also introduce notable burdens, including knowledge acquisition, ontology misalignment, rule conflict, and maintenance overhead.

Conclusions: Decision-maker-governed AI should therefore be understood not as a proven solution, but as a promising governance-oriented design strategy that requires prospective evaluation, fairness auditing, knowledge-lifecycle tooling, transparent institutional oversight, and cross-site implementation studies before its translational potential can be fully assessed.

Public interest summary: Artificial intelligence (AI) is increasingly used in healthcare, but making a good prediction is not the same as making a safe decision. Real clinical, operational, and policy choices depend on rules, guidelines, and lines of accountability that a purely data-driven model does not capture. This review examines a class of AI systems in which doctors, hospital managers, and policymakers actively write and govern the rules that shape how the AI behaves, rather than leaving decisions to a black box. We find these systems are most convincing for well-structured tasks such as checking clinical documentation, following guidelines, and scheduling hospital resources, but that strong real-world evidence is still limited. We argue such systems should be

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seen as a promising, accountability-focused design strategy that now needs careful evaluation in real health services before wide adoption.

Introduction

Artificial Intelligence (AI) is now embedded across the health system, from image interpretation and risk scoring to documentation, workflow management and digital triage [1–6,74–78]. Deep learning and related data-driven methods have produced important gains in specialist services, imaging and electronic health record prediction, and they continue to expand the technical frontier of what health data can reveal [1–3]. Yet in healthcare, AI is not judged by predictive accuracy alone. Clinical decisions, service planning and policy choices all sit inside settings where errors are costly, data distributions shift, and recommendations must remain interpretable to humans who carry legal and ethical responsibility for acting on them [7–13,101–103]. These pressures have revived interest in a long-standing question in medical AI: *should intelligent systems learn primarily from data, or should they reason explicitly with formal knowledge?*

Early medical expert systems demonstrated that symbolic approaches could produce traceable chains of reasoning and encode domain knowledge directly, but they struggled to scale up because they depended on labour-intensive knowledge engineering and on brittle rules, that is, hand-crafted logic that fails when confronted with inputs, exceptions, or contexts its authors did not anticipate [14–16,79–82]. Machine Learning (ML) shifted this balance by learning rich representations directly from data, but at the cost of opacity, vulnerability to dataset shift and weak alignment with guidelines, operational constraints and governance requirements [7–13,17–19]. In high-stakes health settings, neither approach might be sufficient.

Neuro-symbolic AI, understood as the family of methods that combine neural or statistical learning with explicit symbolic representations such as rules, logic, ontologies, or constraints so that the two operate within a single system, together with related forms of informed ML, has emerged as a substantive attempt to bridge that gap. Rather than treating symbolic reasoning and neural learning as rival paradigms, these approaches combine them. Symbolic components can encode guidelines, ontologies, mechanistic assumptions, action constraints or optimisation policies, while neural components learn from noisy, heterogeneous and high-dimensional data [20–24]. In health, this hybrid architecture is attractive because it reframes AI as a decision system rather than a standalone predictor. The relevant question is no longer only whether a model recognises patterns in data, but whether it does so in a manner that is clinically coherent, operationally feasible and institutionally governable.

This review argues that the most important axis of variation in health AI is not the precise technical mechanism, but the degree to which decision-makers actively shape model behaviour. A hybrid system becomes decision-maker-governed when clinicians, managers or policymakers define the rules, constraints or knowledge structures that influence training, inference, validation or escalation. This perspective matters because it shifts attention from explanation as an optional interface feature to formal representation of human judgement inside the decision pipeline itself. This review therefore examines neuro-symbolic health AI through the lens of decision-making, focusing on where explicit human knowledge is inserted, what functions it serves, and how it changes the burden of evidence required for safe deployment.

Methodology

Review design and search strategy

This study was conducted as a critical narrative review informed by a structured literature search, following established guidance for the conduct and reporting of critical and narrative reviews [68,69]. The aim was to critically examine how explicit symbolic and specialist healthcare knowledge is incorporated into AI systems in ways that shape decisions, workflows, and governance. A critical narrative design was appropriate because the literature spans heterogeneous technical traditions with unstable terminology, prioritising conceptual clarity and analytic depth over exhaustive capture. The search strategy combined a structured database search with targeted source identification. Electronic databases searched were EMBASE, PubMed, CINAHL, OECD health policies and data, NHSEED, Web of Science, SCOPUS, SSRN, Cochrane Library, and ProQuest. Supplementary hand searching was conducted across publisher databases, including ScienceDirect, SpringerLink, Sage, and Wiley. A structured search was undertaken using five query families focused on neuro-symbolic AI, informed or knowledge-infused ML, hospital operations and optimisation, trust and adoption, and review literature relevant to healthcare decision-making and management. The literature search of all sources yielded 3,892 unique citations after removing duplicates, of which 97 were selected for critical appraisal and comprehensive reading following a two-stage screening process. Following recommended practice for critical narrative synthesis [68,69], screening was undertaken by the first author, with full-text records, ambiguous cases, and borderline inclusion decisions discussed with co-authors to verify consistent application of the criteria and to reduce single-reviewer bias. In the first stage, titles and abstracts of all 3,892 de-duplicated records were screened against the three analytic questions defined below. Records were included at this stage if they plausibly addressed at least one analytic question, and were excluded where they unambiguously fell outside scope, including purely methodological or algorithmic papers without a healthcare decision context, benchmark-only predictive studies with no reference to workflow, governance, or implementation, non-health applications, editorials and opinion pieces without substantive argument, and non-English records. This stage excluded 3,551 records and retained 341 for full-text appraisal. In the second stage, the 341 remaining records underwent full-text appraisal against the same three analytic questions. Records were excluded at this stage if the full text confirmed that no explicit symbolic, rule-based, ontological, mechanistic, or optimisation layer was present; if any such layer did not materially influence system behaviour; if the study addressed predictive performance in isolation without reference to the decision context; or if the full text was unavailable. This stage excluded 244 records, and 97 sources were retained for in-depth reading and synthesis. All retained sources are cited in the review: references [68] and [69] are methodological references, and the remaining 101 entries in the reference list constitute the retained corpus, so the reference list corresponds exactly to the sources underpinning the synthesis. The selection process is summarised in the PRISMA flow diagram in Fig. 1. Given the narrative rather than systematic design of the review, the emphasis at both stages was on decision relevance and conceptual contribution rather than exhaustive capture. To strengthen coverage of adjacent and emerging literature, the structured search was supplemented by targeted retrieval from the aforementioned databases, as well as backward and forward citation chaining from key reviews and seminal papers, including foundational expert-system and guideline-modelling work [14,15], informed-machine-learning and neuro-symbolic method surveys [20–24], clinical decision support and explainability reviews [4,16,19], and responsible-AI and governance references [8–13]. Authoritative governance and regulatory documents, including World Health Organization guidance and the European Union AI Act, were also included where directly relevant to the review question.

Screening and eligibility criteria

The search was first completed on November 15, 2025, and updated on May 18, 2026, to capture recent literature in this rapidly evolving field; the update identified four additional sources meeting the inclusion criteria, which were incorporated into the synthesis, bringing the total number of retained sources to 101 [70–73]. Sources were eligible for inclusion if they contributed substantively to at least one of three analytic questions: (a) whether a system contained an explicit symbolic, rule-based, ontological, mechanistic, or optimisation layer; (b) whether that layer materially influenced training, inference, validation, recommendation, optimisation, or escalation; and (c) whether the study addressed the decision context in which outputs were intended to be used, including workflow, governance, human oversight, or implementation implications. Studies were excluded if they focused on predictive performance alone without substantive consideration of how outputs were translated into real decisions or organisational action. Since the term neuro-symbolic AI is used inconsistently in healthcare, adjacent literatures were also included where they illuminated the same underlying design problem, including knowledge-infused learning, theory-guided modelling, documentation validation, and constraint-aware operational AI.

Evidence appraisal and synthesis

Throughout the review, sources were interpreted according to both their technical form and their decision relevance. To avoid conflating different forms of evidence, the analysis distinguishes between primary empirical studies, review-level syntheses, and conceptual, or governance sources. It also applies a pragmatic evidence-maturity framing, separating prospective or comparative empirical evidence from retrospective technical studies and from theoretical argument. Where the literature supported only a design rationale or conceptual promise rather than a demonstrated outcome, this was stated explicitly. To derive the conceptual architecture and design patterns reported in this review, each retained source was charted against three attributes: the presence and type of symbolic layer, the point in the decision pathway at which that layer acts, and the role and authorship of the accountable decision-maker. Recurring configurations were then grouped into the five design patterns summarised in Table 2 and the roles summarised in Table 1, and the full charting of all 101 retained sources, 97 from the original search and four from the May 2026 update, is provided as a Supplementary File.

From symbolic and data-driven AI to decision-maker-governed systems

Medical AI began with symbolic reasoning [14,25]. Early expert systems such as MYCIN encoded diagnostic and therapeutic rules directly, making recommendations traceable and interpretable [14]. Later computer-interpretable guideline models extended this tradition by formalising care pathways and decision logic in shareable form [15]. These approaches established a principle that remains central to healthcare AI; recommendations are clinically useful only when they can be linked to accepted reasoning, standards of care, and accountability structures [4,15,16]. The rise of large digital datasets shifted the field towards data-driven learning. Models trained on images, signals, and electronic health records could identify patterns that were difficult to encode manually and, in some narrow tasks, achieved performance comparable to or exceeding that of clinicians [1–3]. Yet these gains came with familiar liabilities. Black-box systems are difficult to interrogate when they fail, and retrospective performance often degrades when populations, measurement processes, or workflows change [7–10]. For this reason, responsible AI in health has increasingly been framed not in terms of benchmark accuracy alone, but in relation to dataset quality, external validation, implementation design, and

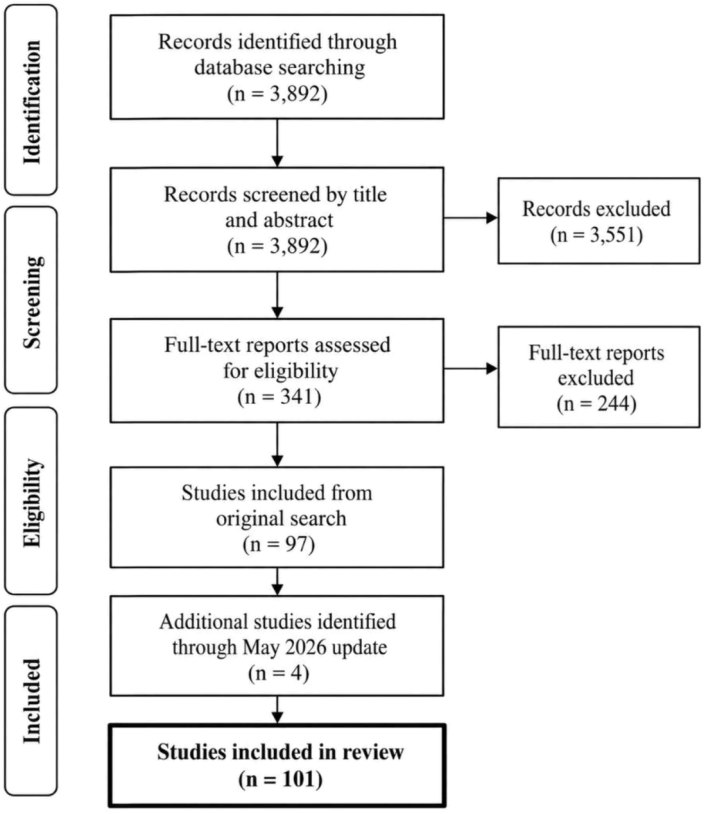


Fig. 1. PRISMA flow diagram showing study selection and inclusion process (total n = 101).

Table 1
Differentiating decision-maker roles in hybrid health AI.

Role	Domain expertise	Basis of authority and accountability	Rate of change and discretion	Typical symbolic contributions
Clinicians/clinical governance groups	Clinical pathophysiology, guideline knowledge, local pathway experience	Professional standards, specialty consensus, medico-legal duty of care	Moderate, with explicit provision for clinical discretion and override	Eligibility criteria, contraindications, escalation thresholds, documentation and handover rules
Managers/service leaders	Service operations, workforce planning, patient flow, organisational constraints	Institutional delegated authority, service performance obligations	Variable with service reconfiguration, limited individual override	Capacity limits, rostering and service policies, overflow and priority rules, scheduling constraints
Policymakers/ regulators	Legal, ethical, and population-health frameworks	Statute, regulation, public accountability and democratic legitimacy	Slow, with minimal scope for local discretion or override	Legal mandates, fairness and equity constraints, protected-group reporting requirements, population-level prioritisation rules

institutional oversight [8–13].

From this perspective, the symbolic versus data-driven divide is less a contest between older and newer paradigms than a question of where knowledge resides within the system. Purely symbolic approaches place knowledge primarily in rules and explicit structures. Purely data-driven approaches place it implicitly in learned parameters. Neuro-symbolic AI and related forms of informed machine learning distribute knowledge across both [20–24]. In technical terms, these systems span a wide range of approaches, from prior knowledge encoded in architectures, loss functions, and features to more explicit combinations of learned representations with logic, programs, knowledge graphs, or formal reasoners [20–24,26]. For healthcare, however, the more useful distinction is operational rather than terminological: what symbolic knowledge is introduced, where it acts in the decision pathway, and whether accountable stakeholders meaningfully author or govern it. This matters because health-system decisions are rarely prediction problems alone. They are shaped by guidelines, action thresholds, escalation logic, resource constraints, and institutional obligations that are not fully captured by statistical learning.

To make this distinction analytically usable, this review defines a system as decision-maker-governed when three conditions are met. First, it contains an explicit symbolic layer. By symbolic layer we mean a human-readable, machine-executable representation of domain knowledge, such as rules, ontologies, graph constraints, mechanistic relations, or optimisation policies, that is expressed in explicit terms rather than learned implicitly as model parameters. Second, that layer materially influences system behaviour during training, inference, validation, recommendation, optimisation, or escalation, rather than functioning only as a descriptive afterthought. Third, accountable decision-makers do more than provide background consultation: they author, approve, adjudicate, or govern the symbolic layer in ways that can change what the system is permitted to output or do [8–13,20–24]. This definition distinguishes hybrid systems in general from the stronger class (of hybrid systems) considered in this review. A model that uses a public ontology for representation learning may be hybrid, but if that ontology is treated as a fixed external resource with no local stakeholder authorship or governance, it is not decision-maker-governed in the sense used here [20,23]. By contrast, a system that embeds clinician-authored validation rules, locally approved escalation logic, or manager-defined operational constraints qualifies because the symbolic layer carries explicit institutional authority. In practice, these three conditions, symbolic layer, behavioural influence, and stakeholder authorship, are jointly necessary. Systems that lack one element may still be technically important, but they should not be treated as straightforward evidence for the stronger governance-oriented category examined here [20,23].

Although this review uses the collective term “decision-maker”, the roles grouped under this label differ in important ways that shape how symbolic knowledge is authored, governed, and revised. Clinicians, and the specialty governance groups in which they sit, typically author symbolic content in the form of eligibility criteria, contraindications, escalation thresholds, and documentation rules that reflect professional standards of care and local clinical consensus. Managers and service leaders encode a different class of symbolic content, including capacity limits, rostering and service policies, overflow rules, and priority structures that translate organisational constraints into admissible actions. Policymakers and regulators shape the symbolic layer from a different level again, through legal mandates, ethical commitments, and population-level prioritisation rules whose authority derives from statute and public accountability rather than from professional judgement alone. These roles therefore differ in *domain expertise, basis of authority, accountability, rate of change, and tolerance for discretion.* Clinical rules change as evidence and local practice evolve and often allow explicit provision for clinician discretion; operational constraints change as services are reconfigured and typically leave less room for individual override; legal and regulatory constraints change more slowly still and are ordinarily not subject to local discretion at all. The decision-maker-governed framing advanced here is intended to accommodate this plurality rather than collapse it: the symbolic layer of a hybrid system will often carry contributions from more than one role, and the governance question is not only who authored it but who is permitted to revise it, on what timescale, and under what conditions. The principal distinctions between these roles are summarised in Table 1.

The conceptual architecture developed across the decision-making framework is summarised in Fig. 2. Three accountable decision-maker constituencies, clinicians, service managers, and policymakers, author and govern an explicit symbolic layer that integrates with a learned component within a single hybrid decision pipeline. Outputs are not surfaced as recommendations until they have traversed explicit governance pathways: validation, constraint, escalation, override, and audit. This schematic is intended not as a prescriptive system blueprint but as an analytic frame that makes visible where institutional knowledge enters the model, where it qualifies model behaviour, and where accountability is preserved.

Across the literature, at least five recurring hybrid design patterns can be identified. Some systems use rule-constrained prediction, in which learned models are regularised or filtered by explicit rules, for example rule-filtered risk models for guideline-concordant triage and alerting [4,16]. Others augment learning with ontologies or knowledge graphs so that predictions and explanations are organised around domain concepts rather than unconstrained features, as in ontology- and knowledge-graph-augmented oncology pipelines that link extracted entities to curated clinical concepts [36,37,83–85]. A third pattern combines neural extraction with symbolic validation, an approach of growing importance in large language model (LLM) pipelines where outputs may be fluent but error-prone, illustrated by symbolic consistency checks applied to neural information extraction from medical reports, where reported neuro-symbolic systems improved the auditability and accuracy of entity recognition and linking from oncologic and radiology notes relative to purely neural baselines [36,37]. A fourth uses mechanistic or theory-guided structures to constrain learning in settings such as physiology or epidemic modelling, for example hybrid mechanistic-ML parameter estimation for infectious disease dynamics [50,51]. A fifth couples prediction with optimisation, embedding learned forecasts within symbolic action layers that encode feasibility,

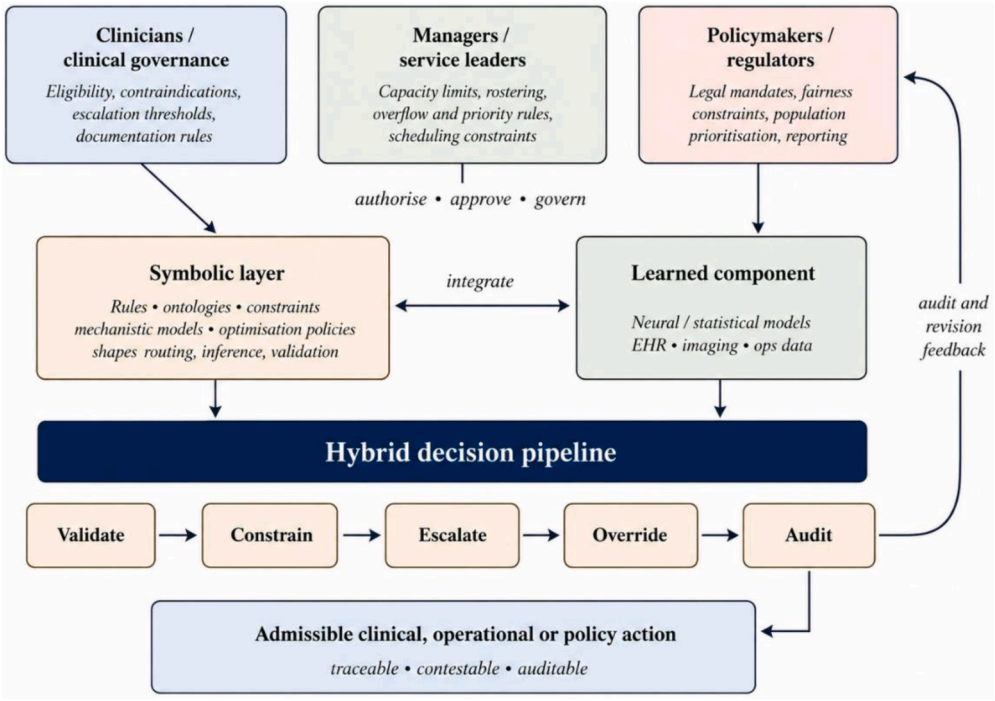


Fig. 2. Conceptual schematic of decision-maker-governed hybrid AI in healthcare.

policy constraints, and institutional priorities, exemplified by answer set programming for operating-room scheduling under bed-management constraints [47–49,94,95]. These patterns are not exhaustive, but they recur consistently enough to provide a useful framework for the evidence reviewed here [4,16,20–24,36,37,40,46–51]. The principal hybrid design patterns identified in the literature are summarised in Table 2, together with their symbolic components, learned elements, and typical healthcare applications.

A critical account must also confront the limitations of symbolic knowledge. Symbolic layers are necessarily incomplete in healthcare, where co-morbidity, workflow variation, and exception states are pervasive [14–16,25]. They revive the knowledge-acquisition bottleneck of earlier expert systems, requiring continuous encoding of local pathways, contraindication rules, and governance decisions. Interoperability remains problematic: ontologies are often treated as portable, yet real systems operate with heterogeneous terminologies and persistent mapping mismatches [27]. Finally, symbolic layers do not remove conflict; they formalise it. Guidelines and fairness commitments may point in different directions, making governance dependent on adjudication and versioning, not representation alone [52,53]. Unresolved normative conflicts within symbolic layers can therefore produce inconsistent outputs across patient subgroups, underscoring the need for explicit adjudication mechanisms and ongoing governance review.

Why healthcare is a strong domain for hybrid decision support

Healthcare is a particularly strong domain for hybrid AI because its decisions are layered, consequential, and institutionally governed [8–13,20–24]. At the bedside, clinicians combine pattern recognition with guideline-based deliberation in a process paralleling dual-process reasoning [28]. At the organisational level, managers translate forecasts into actions constrained by staffing rules, capacity limits, and equity obligations [46]. At the population level, policymakers weigh statistical signals against legal mandates and mechanistic assumptions. Overall, decision quality depends on the interaction between probabilistic inference, formal rules, and accountable human judgement, making health especially fertile ground for hybrid approaches [12,13,50,51].

This is also why purely predictive AI so often falls short. Clinical work unfolds as a chain of linked judgements: whether data are reliable, whether protocol criteria are met, what action is justified, and when responsibility returns to human review. These stages draw on probabilistic, rule-bound, and institutional knowledge. A model may perform well on one stage and still fail if the surrounding decision pathway is not represented [7–10,17]. The attraction of decision-maker-governed design is therefore that it reflects the actual structure of health-system action, determining not only whether patterns exist but when they are sufficient for action, when guidelines apply, and when escalation is

Table 2
Principal design patterns in decision-maker-governed hybrid AI for healthcare.

Pattern	Symbolic component	Learned component	Decision-maker role	Typical health use
Rule-constrained prediction	Clinical rules, thresholds, exclusion criteria	Classifier or risk model	Experts specify admissible recommendations and override points	Diagnosis, triage, alerts
Knowledge-graph or ontology augmentation	Terminologies, ontologies, entity relations	Embeddings, graph models, language models	Experts curate concepts and acceptable reasoning paths	Information extraction, oncology, drug discovery
Neural extraction plus symbolic validation	Consistency checks, report logic, documentation rules	Natural Language Processing (NLP) model or foundation model	Experts define what counts as complete and valid output	Report abstraction, coding, registry capture
Mechanistic or theory-guided hybrid models	Physiological, epidemiological or process models	Residual learning or parameter estimation	Experts specify conserved structure and intervention levers	Forecasting, disease progression, simulation
Prediction plus optimisation	Capacity limits, rostering rules, scheduling policies	Demand or duration prediction	Managers encode feasible actions and institutional priorities	Operating rooms, beds, staffing, patient flow

required [4,14–16,20–24].

Symbolic structure becomes valuable in this setting because it can make those otherwise implicit conditions explicit. It can specify which data elements are required before a recommendation is shown, which findings are incompatible, which co-morbidities alter treatment eligibility, which outputs require senior review, and which documentation elements must be completed before downstream systems can act [4,16,20–24]. In that sense, symbolic knowledge does more than improve interpretability. It can determine whether a model output is sufficiently mature, bounded, and governable to enter the workflow at all. This is also why decision-maker-governed design is more substantively important than generic human-in-the-loop language. A human may be nominally in the loop simply by accepting or rejecting a recommendation after it is produced, and there is reason to believe that this arrangement can give rise to a false sense of security in which the presence of a nominal overseer is mistaken for substantive oversight. In such conditions, critical clinical reasoning may be inadvertently outsourced to the model, with the human reviewer defaulting to acceptance under time and cognitive pressure [7–10,64]. By contrast, decision-maker-governed systems give clinicians, managers, or other accountable actors an upstream role in defining the conditions under which outputs are generated, constrained, validated, escalated, or withheld. The distinction is not merely procedural. It separates systems that ask professionals to absorb model outputs from systems that embed professional judgement within the decision pathway itself [8–13,20–24].

This matters especially under current expectations for responsible health AI. Emerging standards, governance frameworks, and regulatory instruments increasingly converge on a common requirement: health AI must be robust, auditable, subject to human oversight, and deployable in ways that preserve accountability [8–13]. Hybrid systems do not satisfy these conditions automatically, but they make them more tractable by creating explicit sites within the system where constraints, escalation rules, and institutional commitments can be represented, inspected, and revised. That has direct implications for clinical safety. Many harmful failures in healthcare AI arise not because a model output is obviously absurd, but because it is presented too early, too confidently, or without the conditions needed for safe action [7–10,64]. By introducing explicit points of qualification and friction, hybrid systems may reduce these failures even when they do not materially improve predictive accuracy. In a mature healthcare AI ecosystem, that capacity to govern when and how outputs become actionable may prove as important as performance itself.

Clinical applications

The most extensively studied application area for decision-maker-governed hybrid AI is clinical decision support, although the evidence base remains uneven and should not be overstated. In this setting, symbolic layers are most often authored or approved by clinicians and specialty governance groups, whose contributions typically take the form of eligibility criteria, contraindications, escalation thresholds, and documentation rules anchored in professional standards of care. Traditional clinical decision support systems already established why symbolic structure matters in medicine: recommendations must be linked to recognised criteria, not merely to hidden correlations [4,15,16]. Contemporary hybrid systems extend this logic by combining explicit rules, pathways, or validation layers with learned representations derived from unstructured or high-dimensional data. This is important because many clinical decisions are composite rather than singular acts. They require not only prediction, but also interpretation, assessment of eligibility, checking of contraindications, and judgement about whether a recommendation should trigger immediate intervention, deferred review, or specialist escalation. The strongest current support for hybrid design therefore comes from domains in which symbolic validation is already a natural part of the task, particularly documentation and information extraction, oncology workflows, guideline-constrained recommendation, and optimisation under explicit service constraints. By contrast, direct prospective trials in live clinical care remain rare, and many of the strongest claims about trust, fairness, or improved decision quality still rest on retrospective studies, enabling components, or theoretical arguments rather than pragmatic evaluation. This asymmetry is particularly important in a field whose evidentiary standards are grounded in evidence-based practice. The primacy accorded to prospective clinical trials in evidence-based medicine means that retrospective and simulation-based evaluations, however technically sophisticated, cannot on their own carry the weight of a recommendation for clinical adoption. Until hybrid AI systems are tested prospectively, in live care, against clinically meaningful endpoints, claims about their effect on decision quality should be read as hypothesis-generating rather than confirmatory [7–10,64,65].

Imaging, multimodal, and record-based prediction

This distinction is especially clear in imaging, multimodal diagnosis, and electronic health record prediction. Clinical AI gained prominence through perceptual tasks such as image classification, pathology, and risk prediction [1–3,5], but in practice the classifier is rarely the decision itself. The relevant question is whether the output changes diagnosis, treatment, or monitoring. Post hoc explanation alone remains a weak substitute for the reasoning clinicians need [19,29–31]; saliency maps indicate what influenced a model but do not show whether a recommendation is guideline-concordant or whether the threshold for action has been reached. In this setting, symbolic structure serves a different function from explanation. It can encode follow-up rules, required contextual inputs, incompatibilities, and escalation logic, thereby helping determine whether a prediction is actionable within the workflow rather than merely plausible in isolation. This becomes even more important in multimodal systems, where missingness, contradictory evidence, measurement timing, and domain-specific terminology create failure modes that are not resolved by representation learning alone [32]. The same logic applies to temporal prediction from large-scale health records. Deep learning models may identify meaningful patterns in longitudinal data [3,10,33,34], but unless predictions are connected to review intervals, escalation policies, and override conditions, they can increase alert burden without improving care.

Longitudinal, therapeutic, and documentation workflows

Longitudinal monitoring, therapeutic reasoning, and documentation-intensive workflows are particularly revealing because they show where hybrid systems may add practical value without requiring speculative claims about autonomous clinical intelligence. In acute and intensive care, the task extends beyond detecting deterioration to determining justified responses given protocol status and available capacity [7,10,33,35,90–93]. In chronic disease management, repeated observations must be translated into decisions about intensification, contraindications, and referral. In oncology, the symbolic burden is greater still, with treatment decisions depending on staging, biomarker interpretation, and multidisciplinary pathway rules; hybrid extraction and entity-linking systems [36,37] therefore serve as components of a wider clinical reasoning pipeline supporting tumour-board preparation, trial matching, and follow-up planning. Similar logic applies to medication safety, where decisions depend on explicit rules regarding drug interactions, dose adjustment, and review thresholds. In all of these domains, the strongest defensible claim is not yet that hybrid systems improve patient outcomes, but that they can create a better bridge between prediction and accountable action.

Foundation models and generative documentation

The recent literature on foundation models reinforces this point. Large language models and related foundation models appear to encode

substantial clinical knowledge and can perform strongly on benchmark tasks [38,39,86–88], but such performance does not establish safe decision-making in real clinical workflows. Review evidence suggests that these models may support summarisation, retrieval, and draft reasoning while still exhibiting hallucination, unstable calibration, and weak clinical grounding [40]. Comparative work further suggests that even strong general-purpose models can remain substantially weaker than locally trained models for some clinical prediction tasks, with a recent multi-task evaluation reporting that locally trained models outperformed general-purpose large language models across the majority of structured clinical prediction endpoints examined [41]. Recent governance-oriented reviews reach a convergent conclusion, recommending that large language models be confined to bounded, retrieval-grounded, and human-supervised roles rather than autonomous clinical reasoning [70,71,89]. Their translational value therefore lies less in autonomous reasoning than in how effectively they can be bounded through retrieval grounding, symbolic validation, approved templates, and escalation rules. Documentation and digital scribing provide one of the clearest examples. These systems may reduce clerical burden, but they also risk omission, contradiction, and subtle distortion of clinical intent [42–45]. Since documentation shapes coding, follow-up, medication reconciliation, registry capture, and medico-legal accountability, these are not merely linguistic failures. They are decision failures. Hybrid validation is therefore especially compelling here, since it can enforce required elements, detect inconsistencies, and align generated text with institutional schemas and downstream rule-based processes. More broadly, the same reasoning extends to patient-facing communication and shared decision-making. AI-generated summaries, discharge instructions, triage chatbots, and educational tools may appear low risk, but they shape behaviour, escalation, and understanding. In these contexts, symbolic constraints matter because they define what the system is permitted to say, what uncertainty must be disclosed, and when control must return to a clinician.

Across these domains, the central pattern is consistent. Hybrid design appears most credible where decisions are already structured by explicit pathways, formal constraints, or auditable documentation requirements, and where stakeholder-authored rules can plausibly change what the system is allowed to recommend or do. The empirical base is stronger for bounded tasks such as report validation, information extraction, and constrained recommendation than for ambitious claims about bedside reasoning, trust calibration, or patient outcome improvement. That asymmetry should shape how the field is described. The most promising near-term role for decision-maker-governed hybrid AI is not as a replacement for clinical judgement, but as a disciplined framework for making predictions, generated outputs, and recommendations more actionable, contestable, and governable within real healthcare settings.

While clinical decision support has attracted the most research attention, hospital operations and health-system management present arguably clearer structural conditions for hybrid design, because the action space is already defined in explicit institutional and logistical terms.

Operational and managerial applications

Operational and managerial decision-making is, in some respects, an even clearer case for hybrid AI than bedside clinical support. In contrast to the clinical setting considered in the previous section, operational hybrid systems are typically decision-maker-governed by managers and service leaders, whose symbolic contributions encode feasibility, equity, and organisational priorities rather than clinical reasoning per se. Hospital leaders rarely need predictions in isolation. They need decisions about beds, staffing, theatre allocation, discharge prioritisation, and patient flow that remain feasible under explicit organisational constraints [46]. For this reason, healthcare operations research has long combined forecasting with optimisation and constraint management [46]. Although much of this literature sits outside the neuro-symbolic label, it is highly relevant to the present review because it addresses the same underlying problem: how to translate probabilistic estimates into admissible actions under formal rules and institutional priorities. Learned components may estimate surgery duration, admission demand, or discharge probability, while symbolic or optimisation layers convert those forecasts into feasible schedules, allocations, or escalation decisions. In this domain, the symbolic layer is less likely to encode pathophysiology or guidelines than business rules, queueing constraints, rostering policies, fairness requirements, or service-level priorities. That makes operational settings a particularly natural proving ground for hybrid AI, because the action space is already structured in explicit institutional terms.

Patient flow and emergency department management offer another instructive example. Demand for acute beds fluctuates hourly, and decisions about admission, boarding, and discharge interact with staffing, specialty availability, and downstream ward capacity. Predictive models can estimate arrival volumes, length of stay, and discharge readiness, but translating those estimates into actionable bed-management decisions requires explicit encoding of escalation thresholds, overflow policies, and inter-service priority rules. Hybrid designs that couple demand prediction with constraint-aware allocation logic are therefore a natural fit, although most published work remains retrospective or simulation-based [46–49].

The operating-room literature illustrates the mentioned fact clearly. Approaches such as answer set programming have been used to combine scheduling logic with bed-management constraints in ways that make feasibility explicit and inspectable [47], while recent reviews of operating-room scheduling and AI in theatre management point towards increasingly close integration of prediction, optimisation, and human oversight [48,49]. Reported hybrid frameworks that couple machine-learned duration or demand estimates with constraint-based optimisation have described improvements in schedule feasibility and reductions in disruption on retrospective and simulated data, although prospective, multi-site validation in live services remains scarce [48,49,72]. These systems are especially relevant because operational decisions are rarely reducible to technical efficiency alone. They involve negotiated trade-offs across services, workforce fatigue, overflow tolerance, throughput, and strategic priorities, many of which must be encoded directly in the decision machinery. In such settings, explainability is often less about post hoc transparency than about controllability: managers need to know which rule is binding, why a schedule is infeasible, what trade-off a recommendation implies, and how the outcome would change if local priorities were altered. Symbolic optimisation layers are well suited to this form of reasoning because they produce traces that are legible in organisational terms and can often be updated without rebuilding the predictive stack. At the same time, the evidence base remains limited. Many studies report improved schedules or better optimisation on retrospective or simulated data, but far fewer demonstrate sustained gains in throughput, staff experience, or patient outcomes after implementation in live services. The value of the operational literature in this review is therefore twofold: empirically, it broadens the scope of health AI beyond physician-facing systems; conceptually, it shows that decision-maker-governed design is defined not by clinical domain, but by the formal insertion of accountable human constraints into model-guided action.

Population health, fairness, governance and policy

Direct evidence for decision-maker-governed hybrid AI in population health and policy remains limited, but the conceptual case is strong. At the population and policy level, symbolic authorship shifts again: constraints are imposed principally by policymakers and regulators, with symbolic contributions derived from legal mandates, ethical commitments, and requirements of public accountability rather than from professional judgement alone. Public-health and policy decisions

depend not only on forecasting, but also on mechanistic reasoning, legal mandates, ethical commitments, and contested rules of prioritisation. Hybrid machine-learning and mechanistic approaches to infectious disease modelling already demonstrate the value of embedding data-driven inference within explicit system structure, even when they are not framed explicitly as neuro-symbolic systems [50,51,96–98]. Emerging governance frameworks for large-model clinical tools make the same architectural argument explicit, proposing modular, risk-tiered designs in which an updatable knowledge base is useful only when its update process is itself governed [70,73]. The relevance of hybrid design lies less in technical novelty than in governance. As standards for health datasets, World Health Organization (WHO) guidance, and emerging regulation increasingly converge on requirements for robustness, oversight, and accountability, it becomes harder to justify opaque decision systems in high-stakes public settings [11–13]. Questions about who should be prioritised, what trade-offs are acceptable, and when recommendations should be overridden are not purely empirical. They are institutional and normative. Hybrid architectures matter because they create computational sites where such commitments can be represented, inspected, and revised.

Fairness provides the clearest example. Reviews of algorithmic fairness in medicine show that bias mitigation cannot be reduced to statistical post-processing alone [52,53,99,100]. The challenge is already visible in practice: care-management algorithms have reproduced racial inequities when cost is used as a proxy for need [54]; racial bias in pulse oximetry has affected downstream triage and treatment decisions [55]; dermatology AI remains under-validated in populations with skin of colour [56]; and recent benchmarking of large language models suggests that fairness metrics may diverge from clinical usefulness rather than align with it [41]. These cases show both where symbolic structure may help and where its limits remain. Explicit constraints can require subgroup reporting, trigger review when measurement bias is likely, document protected-value trade-offs, or specify different escalation pathways under known conditions of unreliability. They may therefore support fairness auditing and policy transparency. But they do not resolve the normative question of which fairness objective should prevail when metrics conflict, nor can they substitute for legal authority or democratic legitimacy in contested allocation decisions. Population-health applications therefore reveal an important boundary condition. In clinical settings, symbolic structure narrows the action space; in policy settings, it may instead make disagreement explicit, shifting hybrid AI towards traceability and governability of contested choices. This literature remains concentrated in high-income, digitally mature systems, limiting generalisability claims to lower-resource settings.

Positioning the review within prior literature

This review sits at the intersection of several established literatures, but its emphasis differs from each in a way that is analytically important. Broad AI-in-healthcare reviews [5,6,57,58] have mapped opportunities and risks but have generally not treated the formal representation of decision-maker knowledge as a central design problem. Clinical decision support reviews [4,16] have clarified the importance of workflow fit and adoption but rarely examine how hybrid architectures formalise clinician knowledge inside the model. Explainability reviews [19,29–31] have shown why post hoc interpretation is insufficient for high-stakes medical use but remain focused on explaining outputs rather than embedding decision logic inside the system.

Methodological surveys of informed ML and neuro-symbolic AI [20–24] provide deep technical foundations and useful taxonomies, but are mostly domain-agnostic and do not address which knowledge health institutions should formalise, who should govern it, or how such systems should be evaluated in practice. Trust and adoption reviews [59–62] show that clinicians care about autonomy and accountability, but take architecture as given rather than asking how design might make trust more warranted. Foundation model reviews [39,40] suggest medicine is moving towards general-purpose model components but oscillate between excitement and concern about hallucination and weak grounding, without a clear account of how such models should be embedded in health decision systems. The present review addresses that gap.

Taken together, these literatures illuminate important parts of the problem, but they do not fully integrate them. The distinctive contribution of the present review is therefore not simply to introduce neuro-symbolic AI into healthcare, but to reframe the field around the formal role of accountable human knowledge in clinical and health-system decisions. This perspective allows clinical decision support, hospital operations, and policy-facing AI to be examined within a single conceptual frame: hybrid systems are most credible when they encode not only patterns in data, but also the rules, constraints, and institutional commitments under which healthcare decisions are made. The principal points of distinction between this review and major prior review traditions are summarised in Table 3.

Discussion

The central argument emerging from this review is that neuro-symbolic AI is best understood in healthcare not as a narrow technical subfield, but as an alignment architecture for clinical and operational decision-making. Its importance lies in the fact that health decisions are rule-bound, temporally situated, and institutionally accountable. Predictions matter, but they do not by themselves determine what should happen next. In this context, the symbolic layer is not an ornamental addition intended merely to improve explanation. Rather, it is one of the main mechanisms by which learned outputs are translated into admissible action. Fig. 2 summarises this architecture: accountable decision-makers author and govern the symbolic layer, which integrates with a learned component within a hybrid decision pipeline, and outputs are

Table 3
How the present review differs from prior review studies.

Review Type	Representative Literature	Main Contribution	Remaining Limitations
Broad AI-in-healthcare reviews	[5,6,57]	Maps overall field, opportunities and risks	Usually treats reasoning structure and symbolic constraints only briefly
CDSS and implementation reviews	[4,16]	Clarifies workflow fit, adoption barriers and practical deployment concerns	Rarely analyses how hybrid architectures formalise clinician knowledge inside the model
Explainability reviews	[19,29–31]	Shows why post hoc explanation is often insufficient for medicine	Focuses more on interpreting outputs than on constraining decisions through explicit knowledge
Informed ML and neuro-symbolic method surveys	[20–24]	Provides technical taxonomy for knowledge integration	Largely domain-agnostic and weak on institutional governance and clinical workflow
Trust and adoption reviews	[59–62]	Explains why clinicians accept or resist AI tools	Usually treats model architecture as fixed rather than designable for warranted trust
LLM and reasoning reviews	[39,40]	Assesses broad-model potential for clinical reasoning and support	Often lacks a concrete account of symbolic validation, decision rights and action governance

admitted only after traversing explicit pathways for validation, constraint, escalation, override and audit. This framing clarifies what hybrid systems may realistically contribute. They can improve actionability by linking predictions or generated outputs to explicit escalation logic, minimum data requirements, exclusions, and pathways. They can improve knowledge fidelity by giving institutions leverage over model behaviour through explicit representation of guidelines, thresholds, documentation schemas, and operational constraints. They may also create better conditions for calibrated trust by exposing where outputs can be contested, where exceptions enter, and how institutional commitments shape system behaviour [29,59–63]. However, the evidentiary status of these claims is not uniform. The first two (i.e., improve actionability, improve knowledge fidelity) are partially supported in bounded domains such as report validation, information extraction, and constrained decision support. The third remains more inferential and should be treated as a design hypothesis rather than an established outcome. These gains are most visible where symbolic structure is already strong, including guideline-heavy clinical tasks, documentation workflows, scheduling, bed management, and policy-regulated support. In such settings, the question is not whether relevant symbolic knowledge exists, but whether the AI system takes it seriously enough.

At the same time, the rhetoric surrounding hybrid AI can obscure several unresolved problems. Formalising knowledge does not ensure that the knowledge is correct, current, or ethically acceptable. Guidelines may lag behind evidence, local policies may encode historical inequities, and clinician heuristics may themselves be biased or inconsistent. A decision-maker-governed system may therefore crystallise contested judgement rather than resolve it if its symbolic layer is poorly governed. Nor does symbolic structure guarantee genuine interpretability. In many systems the visible rule layer governs only part of the behaviour, or operates only after the main prediction has been generated, creating an appearance of transparency while substantial opacity remains in the learned component beneath it. Claims about interpretability therefore need to be more specific: authors should make clear which parts of the system are constrained, which are inspectable, and which residual behaviours remain opaque. Hybrid design also introduces substantial maintenance burden. Symbolic layers require curation, versioning, adjudication, and retirement. Documentation systems need updated schemas, operational systems need revised constraints as services change, and policy-facing systems require review when legal or ethical mandates shift. In such settings, the main translational risk may move from model development to organisational process. Hybrid AI therefore redistributes labour rather than simply reducing it. These concerns become sharper, not weaker, in the era of foundation models. Their breadth makes them attractive for retrieval, summarisation, and flexible reasoning, but also particularly vulnerable to over-interpretation when fluency is mistaken for warranted competence [38–40]. In this setting, symbolic structure is not optional. It is what can turn broad capability into bounded utility, although even then it does not remove the need for rigorous verification, local governance, and human authorisation.

The review also makes clear that current evaluation standards are inadequate for clinical translation. Many of the most technically interesting hybrid systems are still assessed primarily through retrospective performance, which is insufficient for claims about robustness, actionability, or safety, as demonstrated by the gap between development performance and real-world deployment outcomes [10,64]. Importantly, these limitations cannot be remedied through model-centric evaluation alone. They require study designs that treat symbolic knowledge as a first-class system component, with provenance, revision history, and institutional accountability evaluated *alongside* predictive performance.

A stronger evidence programme is therefore needed, addressing, at minimum, six questions: (1) whether the learned component remains valid across sites and shifting conditions, assessed through external validation and post-deployment monitoring rather than development benchmarks alone [7,10]; (2) whether the symbolic layer functions as intended, with rule conflicts detected, updates versioned, and downstream effects auditable following knowledge revision; (3) whether explanations and constraints improve decisions in use, evaluated through controlled workflow studies examining clinician judgement, override behaviour, and escalation patterns [29,31,60]; (4) whether the system improves workflow outcomes such as review time, alert burden, documentation completeness, or scheduling feasibility, measured through pragmatic service-level trials; (5) whether fairness and accountability pathways are explicit and inspectable, demonstrated through subgroup reporting, governance documentation, and prospective auditing rather than retrospective bias analysis alone [65]; and (6) whether the system can be safely maintained over time, including symbolic knowledge updates, assessed via longitudinal deployment studies rather than static model validation [66].

These dimensions matter especially for bedside support and documentation-intensive use cases, where pilot results can appear compelling while silent errors, coding distortions, handover risks, or degraded practice emerge only at scale [42–45]. More generally, the field needs greater discipline in distinguishing between direct empirical findings, synthesis-level evidence, and conceptual inference. Many published claims slide too easily between these levels. A more mature literature will need prospective, pragmatic evaluation that captures technical and organisational effects.

These observations have implications for researchers, health services, and regulators alike. For researchers, hybrid AI should be studied as a socio-technical arrangement, reporting not only architecture but also symbolic-layer provenance, expert roles, and failure handling. For health services, hybrid AI is not a tool to procure but a capability requiring institutional infrastructure for knowledge authoring, governance, and monitoring. For regulators, symbolically structured systems may be easier to audit, but only when claims about constraint enforcement and oversight are demonstrated rather than assumed. Successful projects will therefore be more multidisciplinary than most current ML efforts [67], with symbolic-layer authorship treated as part of model design from the outset, not an implementation afterthought. Such systems are slower to build but potentially more durable once deployed. The most promising near-term applications are bounded systems that validate documentation, support guideline adherence, constrain operational action spaces, and create better conditions for human review. In healthcare, making the right decision easier, safer, and more traceable is often more valuable than achieving a marginal gain in predictive accuracy alone.

Several limitations of this review should be acknowledged. First, the critical narrative design, while appropriate for synthesising a heterogeneous and terminologically unstable literature, does not provide the exhaustive capture or quantitative synthesis of a systematic review or meta-analysis. Second, the literature reviewed is heavily concentrated in high-income, digitally mature health systems; the extent to which these findings generalise to lower-resource settings with less developed terminology infrastructure and governance capacity remains uncertain. Third, publication bias likely favours positive technical results over implementation failures, meaning that the practical challenges of hybrid deployment may be underrepresented in the evidence base. Finally, the rapid pace of development in foundation models and large language models means that some of the technical landscape described here may have shifted between the final search date and publication.

A broader structural point also warrants explicit acknowledgement. The architectures described in this review are fundamentally dependent on data, whether drawn from electronic clinical records, imaging systems, administrative and operational data, or population-level health information. The benefits of decision-maker-governed hybrid AI are therefore tightly coupled to the availability, quality, and governance of such data. Health systems and countries that lack interoperable records, digital imaging infrastructure, curated terminologies, or population-level health data will not only adopt these technologies more slowly,

but will also find it harder to author and govern the symbolic layers that give hybrid systems their accountability properties. There is a real risk, accordingly, that the AI revolution in healthcare will entrench rather than reduce global inequalities, widening the gap between well-resourced digital health systems and those without comparable data and governance infrastructure. Any serious research and policy agenda for hybrid health AI should treat equitable access to data and infrastructure capacity as a first-order concern, not a downstream consideration [11–13,52–56].

Several priorities follow directly from this review. The decision-maker-governed framing requires independent validation through classification of published systems and examination of how classificatory differences relate to outcomes. Symbolic knowledge lifecycle tooling needs measurement of provenance, update intervals, and the effect of revision on recommendations. Hybrid LLM assistants require comparative workflow trials against unconstrained assistants and conventional decision support [27,41,54–56]. Cross-site and lower-resource studies are needed to establish how dependent this architecture is on strong terminology infrastructure and governance capacity. The field now has enough conceptual clarity to move beyond proof-of-concept demonstrations, particularly in documentation validation, knowledge-grounded decision support, and constraint-aware operations. What remains uncertain is how these systems behave over time inside real institutions with changing workflows, contested rules, and imperfect data. That is the decisive empirical question for the next phase of AI in healthcare.

Conclusion

Neuro-symbolic AI matters in healthcare not because it offers a simple compromise between classical expert systems and modern machine learning, but because it more faithfully reflects the layered structure of clinical, operational, and health-system decision-making. Health services need systems that learn from data while remaining interpretable, rule-constrained, and governable. Decision-maker-governed hybrid models are promising because they create explicit links between pattern recognition, formal knowledge, and human authority. The strongest evidence lies in clinically adjacent applications where symbolic validation is already a natural fit: auditable information extraction, documentation support, guideline-constrained recommendation, and optimisation under operational constraints. Population health and policy applications remain conceptually important but empirically limited.

The key shift required is therefore conceptual as much as technical. Neuro-symbolic AI should be judged not by whether neural and symbolic methods can be combined, but by whether that combination improves the quality, traceability, maintainability, and accountability of healthcare decisions. On that basis, its greatest contribution may be less as autonomous medical intelligence and more as a disciplined framework for clinically grounded, institutionally responsible decision support. The lesson is that healthcare is unlikely to be well served by a simple contest between black-box scale and symbolic purity. A more productive path is selective composition: learned models deployed where data richness matters most, symbolic structures enforced where obligations and constraints matter most, and explicit human authority preserved where value judgement remains irreducible. In this sense, neuro-symbolic AI is not a retreat from modern data-driven systems, but an effort to place machine learning within a more mature theory of clinical action and institutional governance.

Ethics approval

Not applicable. This study is a narrative review of published literature and did not involve human participants, patient data, or animal subjects.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.hlpt.2026.101292.

Data availability

No new datasets were generated or analysed for this review. All source materials used in the manuscript are cited in the reference list.

References

[1] Topol EJ. High-performance medicine: the convergence of human and artificial intelligence. Nat Med 2019;25(1):44–56. https://doi.org/10.1038/s41591-018-0300-7.
[2] Esteva A, Kuprel B, Novoa RA, Ko J, Swetter SM, Blau HM, et al. Dermatologist-level classification of skin cancer with deep neural networks. Nature 2017;542 (7639):115–8. https://doi.org/10.1038/nature21056.
[3] Miotto R, Li L, Kidd BA, Dudley JT. Deep patient: an unsupervised representation to predict the future of patients from the electronic health records. Sci Rep 2016;6 (1):26094. https://doi.org/10.1038/srep26094.
[4] Sutton RT, Pincock D, Baumgart DC, Sadowski DC, Fedorak RN, Kroeker KI. An overview of clinical decision support systems: benefits, risks, and strategies for success. NPJ Digit Med 2020;3(1):17. https://doi.org/10.1038/s41746-020-0221-y.
[5] Jiang F, Jiang Y, Zhi H, Dong Y, Li H, Ma S, et al. Artificial intelligence in healthcare: past, present and future. Stroke Vasc Neurol 2017;2(4):230–43. https://doi.org/10.1136/svn-2017-000101.
[6] Bajwa J, Munir U, Nori A, Williams B. Artificial intelligence in healthcare: transforming the practice of medicine. Future Healthc J 2021;8(2):e188–94. https://doi.org/10.7861/fhj.2021-0095.
[7] Finlayson SG, Subbaswamy A, Singh K, Bowers J, Kupke A, Zittrain J, et al. The Clinician and Dataset Shift in Artificial Intelligence. N Engl J Med 2021;385(3): 283–6. https://doi.org/10.1056/nejmc2104626.
[8] Wiens J, Saria S, Sendak M, Ghassemi M, Liu VX, Doshi-Velez F, et al. Do no harm: a roadmap for responsible machine learning for health care. Nat Med 2019; 25(9):1337–40. https://doi.org/10.1038/s41591-019-0548-6.
[9] Kelly CJ, Karthikesalingam A, Suleyman M, Corrado G, King D. Key challenges for delivering clinical impact with artificial intelligence. BMC Med 2019;17(1):195. https://doi.org/10.1186/s12916-019-1426-2.
[10] Zhang A, Xing L, Zou J, Wu JC. Shifting machine learning for healthcare from development to deployment and from models to data. Nat Biomed Eng 2022;6 (12):1330–45. https://doi.org/10.1038/s41551-022-00898-y.
[11] Arora A, Alderman JE, Palmer J, Ganapathi S, Laws E, McCradden MD, et al. The value of standards for health datasets in artificial intelligence-based applications. Nat Med 2023;29(11):2929–38. https://doi.org/10.1038/s41591-023-02608-w.
[12] World Health Organization. Ethics and governance of artificial intelligence for health: WHO guidance. Geneva: World Health Organization; 2021. ISBN 978-92-4-002920-0.
[13] European Parliament & Council of the European Union. Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence and amending specified Union legislative acts (Artificial Intelligence Act). Off J Eur Union 2024;2024(1689).
[14] Shortliffe EH, Buchanan BG. A model of inexact reasoning in medicine. Math Biosci 1975;23(3–4):351–79. https://doi.org/10.1016/0025-5564(75)90047-4.
[15] Peleg M, Tu S, Bury J, Ciccarese P, Fox J, Greenes RA, et al. Comparing computer-interpretable guideline models: a case-study approach. J Am Med Inform Assoc 2003;10(1):52–68. https://doi.org/10.1197/jamia.m1135.

[16] Montani S, Striani M. Artificial intelligence in clinical decision support: a focused literature survey. Yearb Med Inform 2019;28(01):120–7. https://doi.org/10.1055/s-0039-1677911.
[17] Rudin C. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nat Mach Intell 2019;1(5):206–15. https://doi.org/10.1038/s42256-019-0048-x.
[18] Barredo Arrieta A, Díaz-Rodríguez N, Del Ser J, Bennetot A, Tabik S, Barbado A, et al. Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. Inf Fusion 2020;58:82–115. https://doi.org/10.1016/j.inffus.2019.12.012.
[19] Tjoa E, Guan C. A survey on explainable artificial intelligence (xai): toward medical XAI. IEEE Trans Neural Netw Learn Syst 2021;32(11):4793–813. https://doi.org/10.1109/tnnls.2020.3027314.
[20] von Rueden L, Mayer S, Beckh K, et al. Informed machine learning: A taxonomy and survey of integrating prior knowledge into learning systems. IEEE Trans Knowl Data Eng 2023;35(1):614–33. https://doi.org/10.1109/tkde.2021.3079836.
[21] Karpatne A, Atluri G, Faghmous JH, Steinbach M, Banerjee A, Ganguly A, et al. Theory-guided data science: a new paradigm for scientific discovery from data. IEEE Trans Knowl Data Eng 2017;29(10):2318–31. https://doi.org/10.1109/tkde.2017.2720168.
[22] De Raedt L, Dumančić S, Manhaeve R, Marra G. From statistical relational to neuro-symbolic artificial intelligence. In: In: Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence; 2020. p. 4943–50. https://doi.org/10.24963/ijcai.2020/688.
[23] Dash T, Chitlangia S, Ahuja A, Srinivasan A. A review of some techniques for inclusion of domain-knowledge into deep neural networks. Sci Rep 2022;12(1):1040. https://doi.org/10.1038/s41598-021-04590-0.
[24] Riegel R., Gray A., Luus F., Khan N., Makondo N., Akhalwaya I.Y., et al. Logical Neural Networks. arXiv [Preprint]. 2020; arXiv:2006.13155. doi: https://doi.org10.48550/arXiv.2006.13155.
[25] Miller RA, Pople HE, Myers JD. Internist-i, an experimental computer-based diagnostic consultant for general internal medicine. N Engl J Med 1982;307(8):468–76. https://doi.org/10.1056/NEJM198208193070803.
[26] Ad G, Lamb LC. Neurosymbolic AI: The 3rd Wave. Artif Intell Rev 2023;56(11):12387–406. https://doi.org/10.1007/s10462-023-10448-0.
[27] Fung KW, Xu J, Brear H, Lane A, Lau M, Wong A, D'Havé A. Promoting interoperability between SNOMED CT and ICD-11: lessons learned from the pilot project mapping between SNOMED CT and the ICD-11 Foundation. J Am Med Inform Assoc 2024;31(8):1631–7. https://doi.org/10.1093/jamia/ocae143.
[28] Kahneman D. Thinking, Fast and Slow. New York: Farrar, Straus and Giroux; 2011.
[29] Pierce RL, Van Biesen W, Van Cauwenberge D, Decruyenaere J, Sterckx S. Explainability in medicine in an era of AI-based clinical decision support systems. Front Genet 2022;13:903600. https://doi.org/10.3389/fgene.2022.903600.
[30] van der Velden BH, Kuijf HJ, Gilhuijs KG, Viergever MA. Explainable artificial intelligence (XAI) in deep learning-based medical image analysis. Med Image Anal 2022;79:102470. https://doi.org/10.1016/j.media.2022.102470.
[31] Reddy S. Explainability and artificial intelligence in medicine. Lancet Digit Health 2022;4(4):e214–5. https://doi.org/10.1016/s2589-7500(22)00029-2.
[32] Acosta JN, Falcone GJ, Rajpurkar P, Topol EJ. Multimodal biomedical AI. Nat Med 2022;28(9):1773–84. https://doi.org/10.1038/s41591-022-01981-2.
[33] Rajkomar A, Oren E, Chen K, Dai AM, Hajaj N, Hardt M, et al. Scalable and accurate deep learning with electronic health records. Npj Digit Med (Baltim) 2018;1(1):18. https://doi.org/10.1038/s41746-018-0029-1.
[34] Hripcsak G, Albers DJ. Next-generation phenotyping of electronic health records. J Am Med Inform Assoc 2013;20(1):117–21. https://doi.org/10.1136/amiajnl-2012-001145.
[35] Evans L, Rhodes A, Alhazzani W, Antonelli M, Coopersmith CM, French C, et al. Surviving sepsis campaign: international guidelines for management of sepsis and septic shock 2021. Intensive Care Med 2021;47(11):1181–247. https://doi.org/10.1007/s00134-021-06506-y.
[36] Prenosil GA, Weitzel TK, Bello SC, Mingels C, Manzini G, Meier LP, et al. Neuro-symbolic AI for auditable cognitive information extraction from medical reports. Commun Med (Lond) 2025;5(1):491. https://doi.org/10.1038/s43856-025-01194-x.
[37] García-Barragán Á, Sakor A, Vidal ME, Menasalvas E, Gonzalez JCS, Provencio M, et al. NSSC: a neuro-symbolic AI system for enhancing accuracy of named entity recognition and linking from oncologic clinical notes. Med Biol Eng Comput 2024;63(3):749–72. https://doi.org/10.1007/s11517-024-03227-4.
[38] Moor M, Banerjee O, Abad ZSH, Krumholz HM, Leskovec J, Topol EJ, et al. Foundation models for generalist medical artificial intelligence. Nature 2023;616(7956):259–65. https://doi.org/10.1038/s41586-023-05881-4.
[39] Singhal K, Azizi S, Tu T, Mahdavi SS, Wei J, Chung HW, et al. Large language models encode clinical knowledge. Nature 2023;620(7972):172–80. https://doi.org/10.1038/s41586-023-06291-2.
[40] Mansoor I, Abdullah M, Rizwan MD, Fraz MM. Reasoning with large language models in medicine: a systematic review of techniques, challenges and clinical integration. Health Inf Sci Syst 2025;14(1):6. https://doi.org/10.1007/s13755-025-00403-0.
[41] Brown KE, Yan C, Li Z, Zhang X, Collins BX, Chen Y. Large language models are less effective at clinical prediction tasks than locally trained machine learning models. J Am Med Inform Assoc 2025;32(5):811–22. https://doi.org/10.1093/jamia/ocaf038.
[42] Quiroz JC, Laranjo L, Kocaballi AB, Berkovsky S, Rezazadegan D, Coiera E. Challenges of developing a digital scribe to reduce clinical documentation burden. NPJ Digit Med 2019;2(1):114. https://doi.org/10.1038/s41746-019-0190-1.
[43] van Buchem MM, Boosman H, Bauer MP, Kant IMJ, Cammel SA, Steyerberg EW. The digital scribe in clinical practice: a scoping review and research agenda. NPJ Digit Med 2021;4(1):57. https://doi.org/10.1038/s41746-021-00432-5.
[44] Dymek C, Kim B, Melton GB, Payne TH, Singh H, Hsiao CJ. Building the evidence-base to reduce electronic health record–related clinician burden. J Am Med Inform Assoc 2020;28(5):1057–61. https://doi.org/10.1093/jamia/ocaa238.
[45] Lin SY, Shanafelt TD, Asch SM. Reimagining clinical documentation with artificial intelligence. Mayo Clin Proc 2018;93(5):563–5. https://doi.org/10.1016/j.mayocp.2018.02.016.
[46] Youn S, Geismar HN, Pinedo M. Planning and scheduling in healthcare for better care coordination: current understanding, trending topics, and future opportunities. Prod Oper Manag 2022;31(12):4407–23. https://doi.org/10.1111/poms.13867.
[47] Dodaro C, Galatà G, Khan MK, Maratea M, Porro I. An ASP-based solution for operating room scheduling with beds management. Lect Notes Comput Sci 2019:67–81. https://doi.org/10.1007/978-3-030-31095-0_5.
[48] Al Amin M, Baldacci R, Kayvanfar V. A comprehensive review on operating room scheduling and optimization. Oper Res 2024;25(1):3. https://doi.org/10.1007/s12351-024-00884-z.
[49] Bellini V, Russo M, Domenichetti T, Panizzi M, Allai S, Bignami EG. Artificial intelligence in operating room management. J Med Syst 2024;48(1):19. https://doi.org/10.1007/s10916-024-02038-2.
[50] Jhutty SS, Hernandez-Vargas EA. Parameter estimation in hybrid machine learning and mechanistic models of infectious diseases. IFAC-PapersOnLine 2022;55(16):178–83. https://doi.org/10.1016/j.ifacol.2022.09.020.
[51] Desai AN, Kraemer MUG, Bhatia S, Cori A, Nouvellet P, Herringer M, et al. Real-time epidemic forecasting: challenges and opportunities. Health Secur 2019;17(4):268–75. https://doi.org/10.1089/hs.2019.0022.
[52] Chen RJ, Wang JJ, Williamson DFK, Chen TY, Lipkova J, Lu MY, et al. Algorithmic fairness in artificial intelligence for medicine and healthcare. Nat Biomed Eng 2023;7(6):719–42. https://doi.org/10.1038/s41551-023-01056-8.
[53] Ueda D, Kakinuma T, Fujita S, Kamagata K, Fushimi Y, Ito R, et al. Fairness of artificial intelligence in healthcare: review and recommendations. Jpn J Radiol 2023;42(1):3–15. https://doi.org/10.1007/s11604-023-01474-3.
[54] Obermeyer Z, Powers B, Vogeli C, Mullainathan S. Dissecting racial bias in an algorithm used to manage the health of populations. Science (1979) 2019;366(6464):447–53. https://doi.org/10.1126/science.aax2342.
[55] Sjoding MW, Dickson RP, Iwashyna TJ, Gay SE, Valley TS. Racial bias in pulse oximetry measurement. N Engl J Med 2020;383(25):2477–8. https://doi.org/10.1056/NEJMc2029240.
[56] Liu Y, Primiero CA, Kulkarni V, Soyer HP, Betz-Stablein B. Artificial intelligence for the classification of pigmented skin lesions in populations with skin of color: a systematic review. Dermatology 2023;239(4):499–513. https://doi.org/10.1159/000530225.
[57] Younis HA, Eisa TAE, Nasser M, Sahib TM, Noor AA, Alyasiri OM, et al. A systematic review and meta-analysis of artificial intelligence tools in medicine and healthcare: applications, considerations, limitations, motivation and challenges. Diagnostics 2024;14(1):109. https://doi.org/10.3390/diagnostics14010109.
[58] Yu KH, Beam AL, Kohane IS. Artificial intelligence in healthcare. Nat Biomed Eng 2018;2(10):719–31. https://doi.org/10.1038/s41551-018-0305-z.
[59] Hwang G, Hejazian SS, Vafaei Sadr A, Wagner JK, Hao P, Vemuri A, et al. Translating AI to the bedside with physician buy-in: recommendations from a meta-analysis and systematic review of the literature. Bioengineering 2025;12(12):1363. https://doi.org/10.3390/bioengineering12121363.
[60] Zhang Z, Genc Y, Wang D, Ahsen ME, Fan X. Effect of AI explanations on human perceptions of patient-facing ai-powered healthcare systems. J Med Syst 2021;45(6). https://doi.org/10.1007/s10916-021-01743-6.
[61] Bach TA, Khan A, Hallock H, Beltrão G, Sousa S. A systematic literature review of user trust in ai-enabled systems: an HCI perspective. Int J Hum–Comput Interact 2022;40(5):1251–66. https://doi.org/10.1080/10447318.2022.2138826.
[62] Chew HSJ, Achananuparp P. Perceptions and needs of artificial intelligence in health care to increase adoption: scoping review. J Med Internet Res 2022;24(1):e32939. https://doi.org/10.2196/32939.
[63] Sokol K, Fackler J, Vogt JE. Artificial intelligence should genuinely support clinical reasoning and decision making to bridge the translational gap. NPJ Digit Med 2025;8(1). https://doi.org/10.1038/s41746-025-01725-9.
[64] Sendak MP, Ratliff W, Gao M, Corey K, Havar M, Balu S, et al. Real-world integration of a sepsis deep learning technology into routine clinical care: implementation study. JMIR Med Inform 2020;8(7):e15182. https://doi.org/10.2196/15182.
[65] Collins GS, Moons KGM, Dhiman P, Riley RD, Beam AL, Van Calster B, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. BMJ 2024:e078378. https://doi.org/10.1136/bmj-2023-078378.
[66] Kakarmath S, Esteva A, Arnaout R, Harvey H, Kumar S, Muse E, et al. Best practices for authors of healthcare-related artificial intelligence manuscripts. NPJ Digit Med 2020;3(1):134. https://doi.org/10.1038/s41746-020-00336-w.
[67] Char DS, Shah NH, Magnus D. Implementing machine learning in health care: addressing ethical challenges. N Engl J Med 2018;378(11):981–3. https://doi.org/10.1056/NEJMp1714229.
[68] Grant MJ, Booth A. A typology of reviews: an analysis of 14 review types and associated methodologies. Health Inf Libr J 2009;26(2):91–108. https://doi.org/10.1111/j.1471-1842.2009.00848.x.

[69] Sukhera J. Narrative reviews: flexible, rigorous, and practical. J Grad Med Educ 2022;14(4):414–7. https://doi.org/10.4300/JGME-D-22-00480.1.
[70] Ferreira Santos J, Dores H. Large language models in cardiovascular prevention: a narrative review and governance framework. Diagnostics 2026;16(3):390. https://doi.org/10.3390/diagnostics16030390.
[71] Cao W, Zhang Q, Liu J, Liu S. From agents to governance: essential ai skills for clinicians in the large language model era. J Med Internet Res 2026;28:e86550. https://doi.org/10.2196/86550.
[72] Optimizing operating room scheduling through multi-level learning and column generation: a novel hybrid approach. Health Care Manag Sci 2025;28(4): 691–714. https://doi.org/10.1007/s10729-025-09723-9.
[73] Zhu W, Peng J, Yan Z, Chen Y, Xu J, Zhang L. Designing and evaluating large language model-enabled clinical decision support for heart failure: a modular and risk-tiered framework. Front Digit Health 2026;8:1730457. https://doi.org/10.3389/fdgth.2026.1730457.
[74] Rajkomar A, Dean J, Kohane I. Machine learning in medicine. N Engl J Med 2019; 380(14):1347–58. https://doi.org/10.1056/NEJMra1814259.
[75] Rajpurkar P, Chen E, Banerjee O, Topol EJ. AI in health and medicine. Nat Med 2022;28(1):31–8. https://doi.org/10.1038/s41591-021-01614-0.
[76] Miotto R, Wang F, Wang S, Jiang X, Dudley JT. Deep learning for healthcare: review, opportunities and challenges. Brief Bioinform 2018;19(6):1236–46. https://doi.org/10.1093/bib/bbx044.
[77] Shickel B, Tighe PJ, Bihorac A, Rashidi P. Deep EHR: a survey of recent advances in deep learning techniques for electronic health record analysis. IEEE J Biomed Health Inform 2018;22(5):1589–604. https://doi.org/10.1109/JBHI.2017.2767063.
[78] Xiao C, Choi E, Sun J. Opportunities and challenges in developing deep learning models using electronic health records data: a systematic review. J Am Med Inform Assoc 2018;25(10):1419–28. https://doi.org/10.1093/jamia/ocy068.
[79] Peleg M. Computer-interpretable clinical guidelines: a methodological review. J Biomed Inform 2013;46(4):744–63. https://doi.org/10.1016/j.jbi.2013.06.009.
[80] Shortliffe EH, Sepulveda MJ. Clinical decision support in the era of artificial intelligence. JAMA, J Am Med Assoc 2018;320(21):2199–200. https://doi.org/10.1001/jama.2018.17163.
[81] Bates DW, Kuperman GJ, Wang S, et al. Ten commandments for effective clinical decision support: making the practice of evidence-based medicine a reality. J Am Med Inform Assoc 2003;10(6):523–30. https://doi.org/10.1197/jamia.M1370.
[82] Kawamoto K, Houlihan CA, Balas EA, Lobach DF. Improving clinical practice using clinical decision support systems: a systematic review of trials to identify features critical to success. BMJ 2005;330(7494):765. https://doi.org/10.1136/bmj.38398.500764.8F.
[83] Rotmensch M, Halpern Y, Tlimat A, Horng S, Sontag D. Learning a health knowledge graph from electronic medical records. Sci Rep 2017;7:5994. https://doi.org/10.1038/s41598-017-05778-z.
[84] Choi E, Bahadori MT, Song L, Stewart WF, Sun J. GRAM: graph-based attention model for healthcare representation learning. In: Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining; 2017. p. 787–95. https://doi.org/10.1145/3097983.3098126.
[85] Shang J, Xiao C, Ma T, Li H, Sun J. GAMENet: graph augmented memory networks for recommending medication combination. Proc AAAI Conf Artif Intell 2019;33(1):1126–33. https://doi.org/10.1609/aaai.v33i01.33011126.
[86] Kung TH, Cheatham M, Medenilla A, et al. Performance of ChatGPT on USMLE: potential for AI-assisted medical education using large language models. PLOS Digit Health 2023;2(2):e0000198. https://doi.org/10.1371/journal.pdig.0000198.
[87] Thirunavukarasu AJ, Ting DSJ, Elangovan K, Gutierrez L, Tan TF, Ting DSW. Large language models in medicine. Nat Med 2023;29(8):1930–40. https://doi.org/10.1038/s41591-023-02448-8.
[88] Ayers JW, Poliak A, Dredze M, et al. Comparing physician and artificial intelligence chatbot responses to patient questions posted to a public social media forum. JAMA Intern Med 2023;183(6):589–96. https://doi.org/10.1001/jamainternmed.2023.1838.
[89] Ji Z, Lee N, Frieske R, et al. Survey of hallucination in natural language generation. ACM Comput Surv 2023;55(12):1–38. https://doi.org/10.1145/3571730.
[90] Tomasev N, Glorot X, Rae JW, et al. A clinically applicable approach to continuous prediction of future acute kidney injury. Nature 2019;572(7767): 116–9. https://doi.org/10.1038/s41586-019-1390-1.
[91] Hyland SL, Faltys M, Huser M, et al. Early prediction of circulatory failure in the intensive care unit using machine learning. Nat Med 2020;26(3):364–73. https://doi.org/10.1038/s41591-020-0789-4.
[92] Komorowski M, Celi LA, Badawi O, Gordon AC, Faisal AA. The artificial intelligence clinician learns optimal treatment strategies for sepsis in intensive care. Nat Med 2018;24(11):1716–20. https://doi.org/10.1038/s41591-018-0213-5.
[93] Fleuren LM, Klausch TLT, Zwager CL, et al. Machine learning for the prediction of sepsis: a systematic review and meta-analysis of diagnostic test accuracy. Intensive Care Med 2020;46(3):383–400. https://doi.org/10.1007/s00134-019-05872-y.
[94] Cardoen B, Demeulemeester E, Belien J. Operating room planning and scheduling: a literature review. Eur J Oper Res 2010;201(3):921–32. https://doi.org/10.1016/j.ejor.2009.04.011.
[95] Rais A, Viana A. Operations research in healthcare: a survey. Int Trans Oper Res 2011;18(1):1–31. https://doi.org/10.1111/j.1475-3995.2010.00767.x.
[96] Karniadakis GE, Kevrekidis IG, Lu L, Perdikaris P, Wang S, Yang L. Physics-informed machine learning. Nat Rev Phys 2021;3(6):422–40. https://doi.org/10.1038/s42254-021-00314-5.
[97] Alber M, Buganza Tepole A, Cannon WR, et al. Integrating machine learning and multiscale modeling: perspectives, challenges, and opportunities in the biological, biomedical, and behavioral sciences. NPJ Digit Med 2019;2:115. https://doi.org/10.1038/s41746-019-0193-y.
[98] Willard J, Jia X, Xu S, Steinbach M, Kumar V. Integrating scientific knowledge with machine learning for engineering and environmental systems. ACM Comput Surv 2022;55(4):1–37. https://doi.org/10.1145/3514228.
[99] Gianfrancesco MA, Tamang S, Yazdany J, Schmajuk G. Potential biases in machine learning algorithms using electronic health record data. JAMA Intern Med 2018;178(11):1544–7. https://doi.org/10.1001/jamainternmed.2018.3763.
[100] Chen IY, Pierson E, Rose S, Joshi S, Ferryman K, Ghassemi M. Ethical machine learning in health care. Annu Rev Biomed Data Sci 2021;4:123–44. https://doi.org/10.1146/annurev-biodatasci-092820-114757.
[101] London AJ. Artificial intelligence and black-box medical decisions: accuracy versus explainability. Hastings Cent Rep 2019;49(1):15–21. https://doi.org/10.1002/hast.973.
[102] Ghassemi M, Oakden-Rayner L, Beam AL. The false hope of current approaches to explainable artificial intelligence in health care. Lancet Digit Health 2021;3(11): e745–50. https://doi.org/10.1016/S2589-7500(21)00208-9.
[103] Amann J, Blasimme A, Vayena E, Frey D, Madai VI. Explainability for artificial intelligence in healthcare: a multidisciplinary perspective. BMC Med Inform Decis Mak 2020;20:310. https://doi.org/10.1186/s12911-020-01332-6.